Siqi Guo

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Summary: CMU M.S. ECE graduate with expertise in embedded control, bio-inspired locomotion, and autonomous planning. Currently developing parallelized **C++ IK and motion planning pipelines** for wheeled humanoid warehouse robots at Yondu AI. Previously embedded software engineer at Cepton (LiDAR systems) and research assistant in the CMU Biorobotics Lab (**EigenBot**, **MMPUG**). Published in ICRA 2025, AAAI 2023, and IEEE Neural Networks. Seeking research collaboration in Embodied AI systems, world models for robotics, real-time robot learning, and scalable deployment of foundation models on physical platforms.

Education

Carnegie Mellon University — Pittsburgh, PA
Master of Science in Electrical and Computer Engineering; GPA: 3.91/4.0 — May 2025

Relevant Courses: Foundations of Computer Systems, Embedded System Software Engineering, Machine Learning Signal Processing, Distributed Systems, Autonomous Control System, Storage Systems, Modern Computer Architecture and Design, Large-scale Distributed Machine Learning and Optimization, Introduction to Embedded Systems, Multi-Robot Planning and Coordination, How to Write Fast Code II, Software Development for Computational Science and Engineering

Tianjin University — Tianjin, China
Bachelor of Science in Computer Science and Technology; GPA: 3.84/4.0 — June 2023

Relevant Courses: Data Structure, Operating System, Computer Networks, Principles of Database, Parallel Computing, Natural Language Processing, Pattern Recognition and Deep Learning

Publications

Zhang, Z., **Guo, S.**, et al. “Bio-inspired Distributed Neural Locomotion Controller (D-NLC) for Robust Locomotion and Emergent Behaviors” (May ’25). *2025 IEEE International Conference on Robotics and Automation* (ICRA 2025).

Bingdao, F., Di, J., et al. “Backdoor attacks on unsupervised graph representation learning” (August ’24). *Journal of Neural Networks* (Vol. 180, p.106668).

Jin, D., Feng, B., **Guo, S.**, et al. “Local-Global Defense Against Unsupervised Adversarial Attacks on Graphs” (June ’23). In the *Thirty-Seventh AAAI Conference on Artificial Intelligence* (Vol. 37, No. 7).

Ding, Y., & **Guo, S.** “Conditional generative adversarial networks: Introduction and application” (November ’22). In the 2nd *International Conference on Artificial Intelligence, Automation, and High-Performance Computing* (AIAHPC 2022) (Vol. 12348, pp. 258-266). SPIE.

Research & Work Experience

Yondu AI | Y Combinator, Early-stage Humanoid Robotics Startup — Los Angeles, CA
Robot Intern (Full-time) — Feb 2026 - Present

- Parallelized Motion Planning for Wheeled Humanoid Warehouse Robots
  - Developing parallelized **C++ inverse kinematics (IK) and motion planning pipelines** for wheeled humanoid warehouse robots, targeting real-time performance for autonomous manipulation tasks. Designed **experience-based roadmap cache** system with persistence to accelerate similar pose/trajectory queries, reducing IK solving time for repeated warehouse tasks.
  - Developed a warehouse cart scanning simulation pipeline with QR code localization, IK dispatch for barcode poses, and live virtual camera feedback during scanning.
- Ouster LiDAR Integration & Dynamic IMU Estimation and Calibration System
  - Resolved persistent **asymmetric yaw drift and odometry snaping** in LiDAR-inertial SLAM through: **extrinsic calibration** (KISS-ICP), **EKF sensor fusion audit**, **dynamic IMU bias modeling and correction**, and **LiDAR point cloud deskewing**.
- Data Collection and Post-processing Pipeline Optimization & Acceleration
  - Redesigned cloud-to-dataset post-processing pipeline achieving **5-7x speedup** through CPU/GPU parallelism, breaking through computational bottlenecks and improving throughput from an average of **28 to 145 FPS**.
  - Implemented **parallel segment download** saturating ISP downlink and minimum-coverage data selection for robot training datasets, reducing cloud storage transfer by 92%.

Cepton, Inc. | LiDAR Technology Company — San Jose, CA
Embedded Software Engineer (promoted from Software Engineer Intern, Full-time) — May 2025 - Jan 2026

- MST (Magnosteer Control) System Development
  - Developed embedded **PD control with feedforward** for precision mirror movement control in LiDAR systems, implementing encoder auto-calibration and trajectory interpolation in **C/C++**.
  - Optimized embedded system architecture by redesigning **SPI transfers and event queues**, achieving **73% CPU utilization reduction** (90% → 24%) on ARM-based controllers.

  * Built comprehensive **control simulation framework** in C and Python for algorithm validation, enabling systematic benchmarking of PD vs. PD-with-feedforward control strategies.
  * Engineered real-time **MST GUI and debug logging system** in **Rust**, providing field engineers with live system monitoring and parameter tuning capabilities.
  * Delivered **72% power consumption reduction** (2.9W → 0.8W) on Ultra B3.3 120° LiDAR units through embedded firmware optimization and Python-based power analysis tools.

Eigenbot - Bio-inspired Distributed Control for Modular Robot | Pittsburgh, PA
Biorobotics Lab Research Assistant (Part-time) | Dec 2023 - Oct 2024
Advisor: Professor Howie Choset, Research Scientist Lu Li | Carnegie Mellon University

- Modular Robot Bio-inspired Curve Walking Implementation
  * Designed and implemented a bio-inspired curve walking algorithm in CoppeliaSim for hexapod locomotion, optimizing controller weights for smooth gait transitions and adding steering control into Eigenbot's behavior tree.
- Eigenbot Firmware Development and Testing
  * Developed Eigenbot EEPROM parameters auto-setting tools, which accelerated Hexapod gaits tuning process.
  * Tested the firmware. Redesigned and debugged the module communication protocol to ensure the robot's stability and reliability. Conducted the experiments with Logan, analyzed the gaits and fine-tuned firmware parameters.
  * Integrated Foot Sensor Feedback into the robot's message protocol.
  * Collected EigenBot's data from the real-world using OptiTrack, developing a pipeline to assess metrics like Gait Phase Difference and Leg Stance Duration, benchmarking its performance with centralized predefined gait.

Theory and Applications of Graph Convolutional Network | Tianjin, China
Research Assistant (Part-time) | Nov 2021 - May 2023
Advisor: Associate Professor Jin, Di | Tianjin University

- Investigated the theoretical relationship between GAT and GCN within Deep Graph Infomax (DGI); integrated mutual information maximization and unsupervised learning to improve graph representation quality.
- Benchmarked adversarial robustness of GNNs using Metattack and Nettack attack matrices; reproduced and compared defense methods including RoSA, Pro-GNN, PA-GNN, GraphCL, and GraphSS.
- Designed a novel **unsupervised backdoor attack** on GNNs based on GIN and Momentum Contrast, formulated the threat model, derived the loss function, and implemented trigger injection; achieved higher Attack Success Rate and lower Clean Accuracy Drop than prior unsupervised graph attacks.

Projects

CloudFS, a Deduplicated Cloud File System | CMU 18-746 Course Project | Oct 2024 - Dec 2024

- Implemented CloudFS, leveraging the properties of local SSD and cloud storage to make data-placement decisions.
- Identified and eliminated duplicate data with Rabin fingerprinting. Minimized storage usage and optimized data transfer in the form of segments. Designed buffer file to manually handle the file data transfer.
- Implemented the IOCTL functions to support snapshot operations, such as create, restore, install, and uninstall.
- Added in-memory LRU caching to further improve performance and reduce cloud operation costs, reaching **NO.1** on the scoreboard among all students.

BusTub, a Relational DBMS | Personal Project based on CMU 15-645 | May 2024 - July 2024

- Implemented BusTub DBMS storage layer (LRU-K, disk scheduler, buffer pool, extendible hash index), query execution (operator executors, optimizer rules), and transaction support with optimistic multi-version concurrency control.

Distributed File System Design | CMU 15-640 Course Project | Feb 2024 - Apr 2024

- Built a File-Caching Proxy over RMI with open-close session semantics, LRU cache replacement, and version-based freshness control; implemented low-level RPC serialization for NFS operations.
- Designed a Two-phase Commit Protocol with failure recovery and lost-message handling; tuned a 3-tier auto-scaling web service and benchmarked system bottlenecks under dynamic load.

Blind Source Signal Decomposition & Analysis | CMU 18797 Research Project | Oct 2023 - Dec 2023

- Filtered the mixed HDEMG signals and detected peaks to slice the peak-wave windows. Used the PCA algorithm to reduce the dimensions of the data and K-Means to cluster the potentially same Motor Units' signals.
- Based on the clustered results, applied DTW, Covariance, Cosine in a multi-threshold way to compare the wave similarity. Verified the firing instances for each Motor Unit based on the comparison results.

Skills

Programming: C/C++, Python, Rust, LaTeX, Verilog
Tools: GIT, Docker, MySQL, VSCode, Gem5, PSoC, Vitis
Frameworks: ROS, TensorFlow/PyTorch, Isaac Sim, Drake
Platforms: MacOS, Linux, Arduino, GCP